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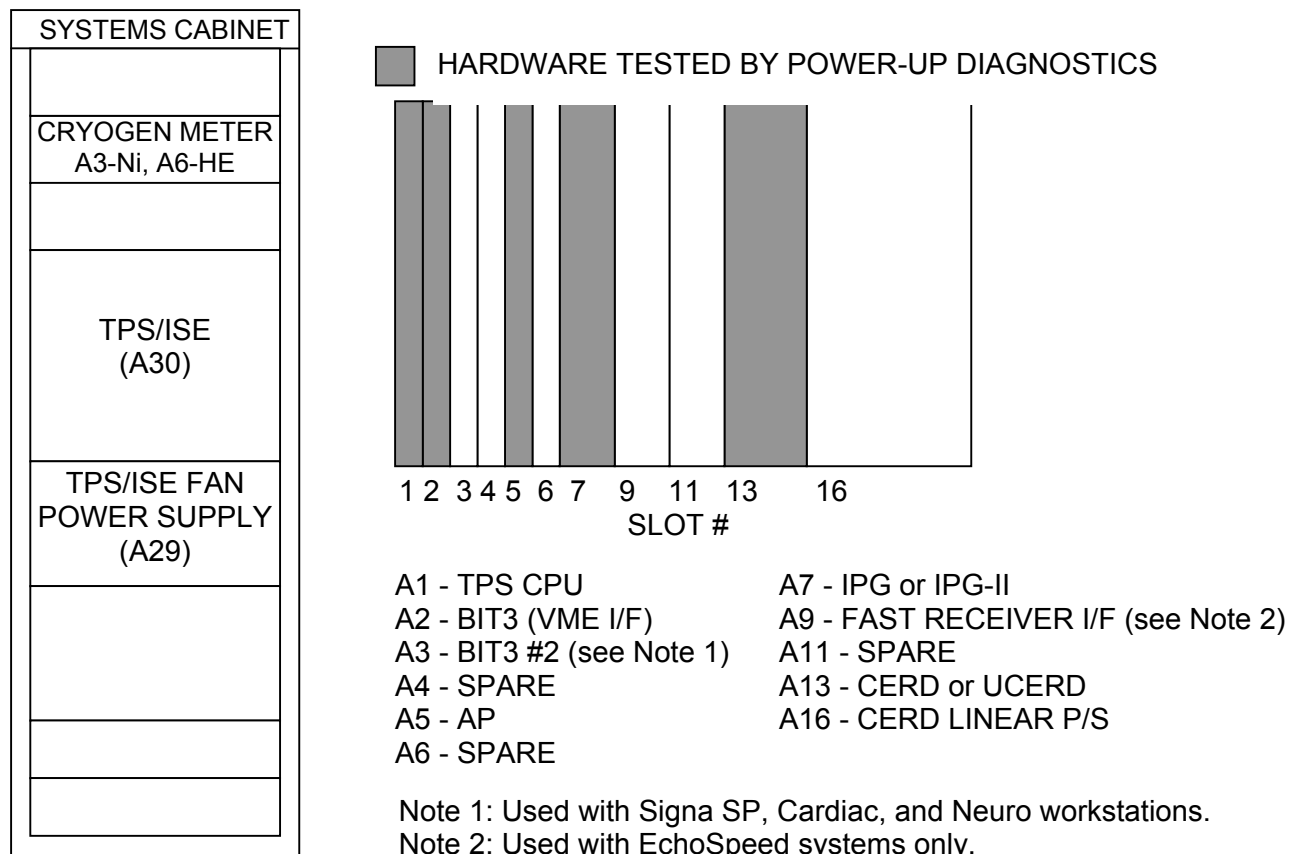
Description - This procedure explains how to start TPS/ISE power-up tests, what test LEDs are provided, and test descriptions.

Note

Do not start multiple copies of diagnostics. Multiple copies of diagnostics have caused different anomalies such as locked up diagnostic windows.

1- HARDWARE TESTED

Power-up diagnostics (EPROM-based) check only a portion of the TPS/ISE subsystem hardware (refer to Illustration 1-1). The power-up diagnostics, at a minimum, test all subsystem hardware required for establishing communications with the host computer.



TPS/ISE TESTED HARDWARE LOCATION
ILLUSTRATION 1-1

2- STARTING TPS/ISE POWER-UP TESTS

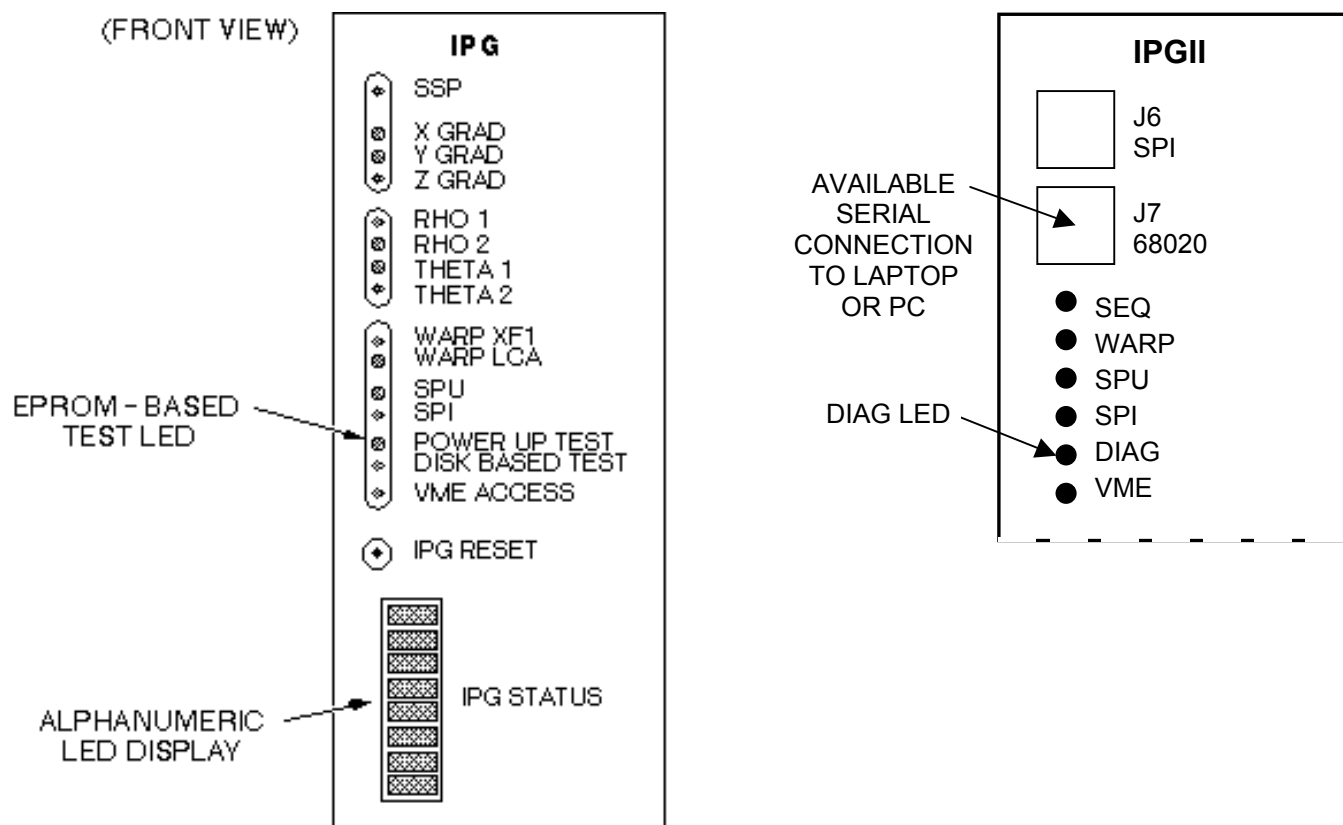
EPROM-based tests, which take ~1:40 seconds, execute when TPS/ISE subsystem power is turned ON, or the TPS/ISE is reset. A TPS/ISE reset occurs via software (either **[TPS Reset]**, or when TPS/ISE diagnostics are selected/run from the Utility Menu). Power-up test errors are reported on the alphanumeric LED display (IPG) or via the serial connection to a laptop computer (IPG-II); refer to Section 3 - TPS/ISE Test Indicators. An attempt is also made to report pass/fail status back to the message log.

3- TPS/ISE TEST INDICATORS

Local test indicators are provided on the Integrated Pulse Generator (IPG, IPG-II) Board. These LEDs (see Illustration 3-1) provide backup error/status indication for TPS/ISE tests in case communications with the host computer fails.

IPG: One green LED (POWER UP TEST) and an alphanumeric LED display provide the status of TPS/ISE EPROM-based tests and general subsystem activity. First, all LEDs on the IPG are flashed to verify that they are working. When the IPG starts power-up testing, all LEDs remain on except the SPI, DISK BASED TEST and VME ACCESS LEDs. During the power-up test sequence, the IPG STATUS displays the abbreviated names of the tests while they are running. "PUD PASS" appears when the power-up tests are complete. If the IPG fails any fatal power-up test, testing halts and the IPG STATUS displays the numeric error code. Any nonfatal error codes are also displayed, but testing continues.

IPG-II: One green LED (DIAG) and a serial connection (to laptop or PC) provides an indication of power-up tests. (Refer to procedure "TPS/ISE Serial Debug Connection"; *this proprietary procedure is available for GE use, and to sites with a valid Advanced Service Package Limited License.*) First, all LEDs on the IPG are flashed to verify that they are working. When the IPG-II starts power-up testing, all LEDs remain on except the SEQ and WARP. On the IPG-II, TEST PASSED appears on the laptop/PC display via the serial connection as each test is completed. If the IPG-II fails any fatal power-up test, testing halts and the laptop/PC displays the numeric error code. Any nonfatal error codes are also displayed, but testing continues.



IPG TEST INDICATORS
ILLUSTRATION 3-1

4- TPS/ISE POWER-UP TEST DESCRIPTIONS

4-1 Integrated Pulse Generator (IPG/IPG-II)

IPG power-up tests verify the integrity of hardware required for the IPG to boot via the network. Total test time is ~1:40 seconds. If a fatal error occurs, diagnostics are halted. If a nonfatal error occurs, the diagnostics report the error and continue diagnostic processing. Refer to the following test descriptions

- **EPROM Test (Fatal)** - Test time is ~1 second. This test verifies the integrity of the data in the two 128KB EPROMs (i.e. that the data hasn't changed since they were programmed). The computed checksum of each EPROM is verified against the embedded checksum.
- **Local Bus Test (Fatal)** - Test time is ~ 0.1 second. This test verifies that each bit can be set and reset on the local bus. A set of local bus patterns are written to the 32-bit diagnostic register and read back. A failure occurs if the value read back is not the inverse value just written. Patterns are used in this order: 55555555, AAAAAAAAAA, 33333333, 0F0F0F0F, 00FF00FF, 0000FFFF.
- **DRAM Test (Fatal)** - Test time is ~ 16 seconds. This test verifies overall DRAM operation. More extensive tests are run in the disk-based diagnostics to thoroughly verify DRAM operation. The DRAM test is composed of three sub-tests: Memory Cell Test, Memory Refresh Test, and Memory Addressing Test.
 - Pattern Test - This test verifies each memory bit can be set and cleared and no bit-to-bit shorts exist. Patterns used are: 55555555, AAAAAAAAAA, 33333333, 0F0F0F0F, 00FF00FF, 0000FFFF.
 - Refresh Test - This test is performed to verify that the hardware refresh circuitry is functioning properly. After the memory cell test finishes, there is a two second pause before performing the memory refresh test. This test reads back all of the tested memory (64KB) and verifies that the last written value, 0000FFFF, is still present in each location.
 - Address Test - This test verifies that the address lines are functioning properly. The test begins by writing 32-bit, long word addresses into the long words specified by their addresses. The test then verifies that each long word location contains its address as data. The test will fail if any long word contains a value other than its address.
- **Multifunction Peripheral (MFP) Test (Fatal except for serial controller)** - Test time is ~ 0.1 second. This test verifies that the MFP device is functioning properly. There are two MFPs provided on the IPG board. MFP chip 1 is not tested since it is not used during an IPG boot. Only MFP chip 2 is tested during power-up. This test is made up of four sub-tests: Register Test, Timer Test, Timer Interrupt Test, Serial Communication Controller Test.

The MFP registers are tested by writing and reading appropriate bit patterns to verify that each bit can be set and reset. The MFP timer test verifies the four internal timers in the MFP as follows:

- **Timer Linearity Test** - Each timer is programmed to begin counting down from F0. It is then repeatedly examined to ensure the count is monotonically decreasing and that it resets to F0 and continues counting.
- **Timer Prescaler Test** - Timer A is tested with respect to timers B, C, and D. Timer B is individually tested with respect to timer A. The timer pairs are tested using different sets of prescaler values and compared with each other. Any discrepancy is reported as an error.
- **Timer Interrupt Test** - Timer A and B are set to Delay Mode with a prescaler value of 200. The initial count for timer A is set to 2; the initial count for timer B is set to 1. Both counters then count down and generate interrupts on a count of 0. Each counter should reload automatically and repeat the count down. After a 5 msec pause, each counter is checked to verify it generated the expected number of interrupts. Counters C and D are then similarly checked.

The MFP serial controller channel is tested at 9600 baud using an internal loopback as follows:

- **Transmit/Receive Data Test** - All interrupts are disabled. After checking for transmit buffer empty, a character is output. After a 2 msec pause, the test checks for receiver full status, reads the input character, and compare the data read with the data transmitted. Next, the test checks for transmit buffer empty, outputs two characters, and then checks immediately to verify the transmit buffer is no longer empty.
- **Transmit/Receive Interrupts Test** - Interrupts are enabled for both transmit and receive character events. A character is placed in the transmit register. If both the transmit and receive interrupts did not occur within 2 msec, an interrupt error is reported.
- **Floating Point Coprocessor Test (Fatal)** - Test time is ~ 0.01 seconds. This test verifies the FPP device is installed and functioning correctly. The following set of tests is performed:
 - **Floating Point Installed Test** - The FPP is requested to perform a floating point instruction. If a bus error is not generated, then the coprocessor is considered to be installed.
 - **Floating Point Data Register Test** - The eight floating-point data registers in the FPP are tested using a 32 bit sliding 1's pattern.
 - **Floating Point Multiply Test** - The floating point register 0 (FP0) is loaded with 2 and FP1 is loaded with 6. The instruction "FMUL.P FP0, FP1" is executed and the product in FP1 verified.
 - **Floating Point Sine Test** - The floating point register 1 (FP1) is loaded with 12. The instruction "FSIN.P FP1" is executed and the result in FP1 is verified.

- Floating Point Add Test - The floating point register 0 (FP0) is loaded with 5 and FP1 is loaded with 8. The instruction "FADD.P FP0, FP1" is executed and the sum in FP1 is verified.
- Floating Point Subtract Test - The floating point register 0 (FP0) is loaded with 21 and FP1 is loaded with 32. The instruction "FSUB.P FP0, FP1" is executed and the difference in FP1 is verified.
- **Non-Volatile RAM Test (Fatal)** - Test time is ~ 1 second. In this test, the contents of the NVRAM is saved in the DRAM. Then it is verified that each bit in the NVRAM can be set and reset. The test also verifies that every NVRAM location can be addressed properly.
- **VME chip Test (Fatal)** - Test time is ~ 1 second. This test verifies the operation of the VME chip from the local side. Pattern tests (55, AA, 33, OF) are run on the following VME chip registers: Board Identification Register and General Purpose Registers CSR0 through CSR4.
- **IPG to CPU Interprocessor (Fatal)** - Test time is ~ 1 second. This test verifies the integrity of the communication paths between the TPS/ISE CPU board and the IPG board.
 - IPG -> CPU Memory Test - The CPU VMEbus interface is set up as the bus slave and the IPG VME bus interface is set up as a bus master. The memory test for a block of TPS/ISE CPU RAM (100000H to 3FFFFFFH) is performed by the IPG board.
 - IPG -> CPU Pattern Test - Long word pattern tests are performed to verify 32-bit data transfer capability. The patterns are written to a location of the TPS/ISE CPU and read back. A failure occurs if the value read back does not match the value written.
 - IPG -> CPU Address Test - This test verifies the address lines are functioning properly. Long word (32-bit) addresses are written into the long words specified by the addresses. These long word locations are verified that they contain all the addresses as data. A failure occurs if any long word contains a value other than its address.
 - IPG -> CPU Interrupt Test - The TPS/ISE CPU prepares to receive an interrupt from the IPG. If it does not receive the interrupt, a failure occurs.

4-2 CPU Board

Total test time is ~22 seconds. Fatal tests halt the power-up diagnostics; nonfatal tests report the error to the Signa message log, but do not halt the diagnostics. Refer to the following test descriptions:

- **EPROM (Fatal)** - Test time is ~ 0.1 seconds. This test verifies the integrity of the data in the two or four 64KB EPROMs (as applicable). The computed checksum of all EPROMs is verified against the embedded checksum.
- **RAM (Fatal)** - Test time is ~ 15 seconds. This test verifies the 4MB of CPU Board DRAM as follows:
 - Memory Cell Test - This test verifies each memory bit can be set and cleared. Patterns used are: 5555555H, AAAAAAAAAH, 33333333H, 0F0F0F0FH, 00FF00FFH, and 0000FFFFH.
 - Refresh Test - This test is then performed to verify that the hardware refresh circuitry is functioning properly. The approximate refresh cycle time for the entire DRAM is 8 msec. For this test, the CPU waits for at least 2 sec before accessing the DRAM, then the last written data from the memory cell test (0000FFFFH) is read back again from each location and verified.
 - Address Test - This test verifies that all address lines are functioning properly, byte/word control lines are functioning correctly, and all memory locations are addressable. Each 32 bit (long word) memory location is written with the hex value of its memory address (e.g. address 3F0000H is written with long word 3F0000H). Each location is then read back and verified for correct data.
- **Serial Controller (Non-fatal)** - Test time is < 0.1 seconds. This test verifies the two serial channels on the Z8530 Serial Controller. Since neither of these channels is used for normal application software, the test is "non-fatal". The functionality tested is the ability to transmit and receive characters, the ability to interrupt when a character is received, and the ability to interrupt after a character has been transmitted. The Z8530 contains an internal loopback which enables the transmit lines to be connected to the receive lines for the tests. Testing is at 9600 baud.
 - Transmit subtest #1 - All interrupts are disabled. A character is then written to the appropriate channel. After a 2 msec pause, the character is read back from the channel. If there is no character available in the receive buffer, or if the character read in is not the same as the one written, an error is reported.
 - Transmit subtest #2 - Interrupts are enabled only for receive character events. A character is placed in the transmit register. If the receive interrupt does not occur within 2 msec, an interrupt error is reported.
 - Transmit subtest #3 - Interrupts are enabled only for transmitted characters (transmit buffer empty). A character is placed in the transmit register. If the transmit interrupt does not occur within 2 msec, an interrupt error is reported.

- **Multifunction Peripheral (MFP) Test (Fatal except for serial controller)** - Test time is ~ 0.1 second. This test verifies that the MFP device is functioning properly. There are two MFP's provided on the IPG board. MFP chip 1 is not tested since it is not used during an IPG boot. Only MFP chip 2 is tested during power-up. This test is made up of four sub-tests: Register Test, Timer Test, Timer Interrupt Test, Serial Communication Controller Test.

The MFP registers are tested by writing and reading appropriate bit patterns to verify that each bit can be set and reset. The MFP timer test verifies the four internal timers in the MFP as follows:

- **Timer Linearity Test** - Each timer is programmed to begin counting down from F0. It is then repeatedly examined to ensure the count is monotonically decreasing and that it resets to F0 and continues counting.
- **Timer Prescaler Test** - Timer A is tested with respect to timers B, C, and D. Timer B is individually tested with respect to timer A. The timer pairs are tested using different sets of prescaler values and compared with each other. Any discrepancy is reported as an error.
- **Timer Interrupt Test** - Timer A and B are set to Delay Mode with a prescaler value of 200. The initial count for timer A is set to 2; the initial count for timer B is set to 1. Both counters then count down and generate interrupts on a count of 0. Each counter should reload automatically and repeat the count down. After a 5 msec pause, each counter is checked to verify it generated the expected number of interrupts. Counters C and D are then similarly checked.

The MFP serial controller channel is tested at 9600 baud using an internal loopback as follows:

- **Transmit/Receive Data Test** - All interrupts are disabled. After checking for transmit buffer empty, a character is output. After a 2 msec pause, the test checks for receiver full status, reads the input character, and compare the data read with the data transmitted. Next, the test checks for transmit buffer empty, outputs two characters, and then checks immediately to verify the transmit buffer is no longer empty.
- **Transmit/Receive Interrupts Test** - Interrupts are enabled for both transmit and receive character events. A character is placed in the transmit register. If both the transmit and receive interrupts did not occur within 2 msec, an interrupt error is reported.
- **Real-Time Clock (Non-fatal)** - Test time is ~ 5 seconds. This test checks for a good battery status, verifies that the clock counters are advancing, and verifies that all elements of the clock will roll over at the proper value and increment the next highest order counter. This last check is made by setting the clock to a value of 23 hours, 59 minutes, 59 seconds, and waiting for 1.1 seconds. A check is then made to verify that all clock counters have rolled over.

- **Floating Point Coprocessor Test (Fatal)** - Test time is ~ 0.01 seconds. This test verifies the FPP device is installed and functioning correctly. The following set of tests is performed:
 - Floating Point Installed Test - The FPP is requested to perform a floating point instruction. If a bus error is not generated, then the coprocessor is considered to be installed.
 - Floating Point Data Register Test - The eight floating-point data registers in the FPP are tested using a 32 bit sliding 1's pattern.
 - Floating Point Multiply Test - The floating point register 0 (FP0) is loaded with 2 and FP1 is loaded with 6. The instruction "FMUL.P FP0, FP1" is executed and the product in FP1 verified.
 - Floating Point Sine Test - The floating point register 1 (FP1) is loaded with 12. The instruction "FSIN.P FP1" is executed and the result in FP1 is verified.
 - Floating Point Add Test - The floating point register 0 (FP0) is loaded with 5 and FP1 is loaded with 8. The instruction "FADD.P FP0, FP1" is executed and the sum in FP1 is verified.
 - Floating Point Subtract Test - The floating point register 0 (FP0) is loaded with 21 and FP1 is loaded with 32. The instruction "FSUB.P FP0, FP1" is executed and the difference in FP1 is verified.

CPU to IPG Interprocessor (Fatal) - Test time is ~ 1 second. This test verifies the integrity of the communication paths between the TPS/ISE CPU board and the IPG board. The following set of tests are similar to the tests completed by the IPG Power Up Test Sequence:

- CPU -> IPG Memory Test - The CPU VMEbus interface is set up as the bus master and the IPG VME bus interface is set up as a bus slave. The memory test for a block of IPG RAM (900000 to BFFFFFFH) is performed by the TPS/ISE CPU board.
- CPU -> IPG Pattern Test - Byte pattern, short word pattern, and long word pattern tests are performed to verify 8, 16, and 32-bit data transfer capability. The patterns are written to a location of the IPG board and read back. A failure occurs if the value read back does not match the value written.
- CPU -> IPG Address Test - This test verifies the address lines are functioning properly. Long word (32-bit) addresses are written into the long words specified by the addresses. These long word locations are verified that they contain all the addresses as data. A failure occurs if any long word contains a value other than its address.

- CPU - > IPG Interrupt Test - The IPG prepares to receive an interrupt from the TPS/ISE CPU. If it does not receive the interrupt, a failure occurs.

4-3 Bit3, AP, and CERD Boards

The Bit3, AP, and CERD Boards have their own powerup tests, but there are no local display indications for pass/fail status of these tests.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 29, 1998	J. Saperstein	Initial change from Toolbook to Word.
1	Oct. 14, 1999	M. Keber	Corrections to TPS Hardware tested illustration; added IPG-II.